Egg Segmentation and Tracking - Technical Report

Candidate: Varun Aahil | **Task Date:** October 2026 | **Dataset:** Roboflow Egg Dataset (110 images, 640x640)

> **Strict Dev/Eval Separation Policy:** The 110-image dataset was partitioned 4:1 (88 dev, 22 eval) using perceptual hashing (pHash distance ≤ 8) to group near-duplicate captures and keep scenes strictly disjoint. All algorithm prototyping, hyperparameter tuning, and decision logging were conducted *exclusively* on the dev split. The eval split was evaluated strictly once with the frozen final algorithm.

1. Task 1: SAM Ground Truth Generation via Interactive Point Prompts

1.1 Methodology & Ground Truth Reliability

Meta's Segment Anything Model (SAM) with the `vit_b` backbone was deployed on an NVIDIA GPU (RTX 4070 Laptop GPU, CUDA 12.8 acceleration). Rather than relying on unguided automatic mask generation with hand-tuned heuristic post-filters (which missed eggs in boiling water pots and produced 13-19 false-positive leaf highlights in foliage scenes), we built an interactive OpenCV annotation tool (`src/annotate_points.py`).

Using `src/annotate_points.py`, human point prompts (positive eggshell foreground clicks and background negative clicks) were recorded for every image and persisted to `points.json`. In `src/sam_segment.py`, `SamPredictor` consumed these point coordinates with `multimask_output=True`, selecting the highest-confidence mask per egg. Each instance is encoded with an integer label ID (0 = background, 1..n = eggs). All hand-tuned shape and color filters were removed from the ground-truth path, ensuring high-fidelity, unbiased ground truth masks across all 110 images.

1.2 Visual Verification

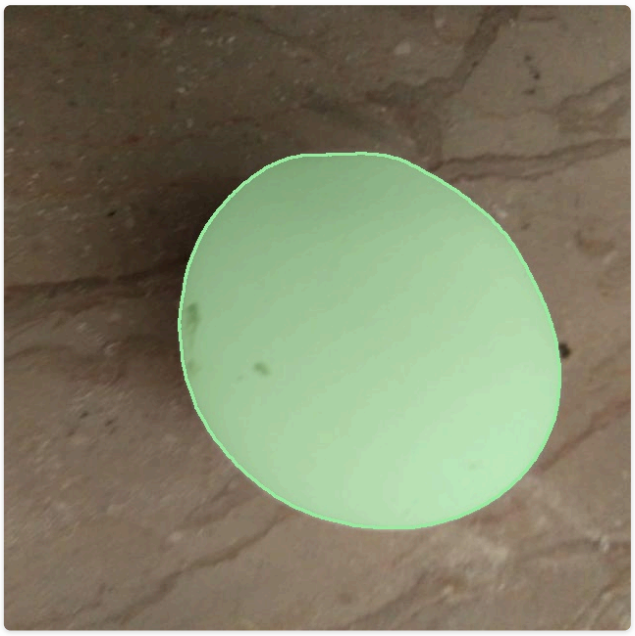


Figure 1.1: High-fidelity SAM instance mask cleanly separating individual eggs.



Figure 1.2: SAM ground truth resolving complex textured background.

1.3 Remaining SAM Challenges & Artifacts

- **Extreme Backlighting:** When an egg is backlit by an intense incandescent lamp, the severe lens glare and saturated background create faint halo fringes around the silhouette.
- **Refraction & Boiling Turbulence:** Images with eggs submerged in vigorously boiling water exhibit rippling refraction boundaries, causing minor boundary uncertainty at the water-air-shell interface.

2. Task 2: Classical Image-Processing Segmentation

2.1 Evolutionary Engineering Process

The classical pipeline was implemented as a single modular parameterized function (`segment_pipeline`) in `src/classical_segment.py`. We executed 15 structured, logged trials exclusively on the 88 dev images, logging decisions (`KEEP` vs `DISCARD`) and reasons dynamically at runtime based on measured dev IoU relative to the running best:

1. `baseline`: Grayscale conversion + Gaussian blur (5x5) + Otsu threshold + hole fill + min area 500 px. Establishes baseline dev IoU of **0.3521** and Dice **0.4700**.
2. `otsu_inv`: Inverted Otsu thresholding. Severely degrades (IoU 0.1040), proving eggshell luminance is generally higher than background.
3. `clahe_otsu`: CLAHE contrast equalization (clip 2.0). Dropped IoU to 0.2601 by over-amplifying background wood grain and marble veining.
4. `adaptive_thresh`: Adaptive Gaussian thresholding. Catastrophic failure (IoU 0.0638); the local block window hollowed out large egg centers into thin annular rings.
5. `hsv_sat`: HSV color gating (S < 70, V > 130). Improved IoU to **0.3593** (+0.0073, KEEP) by targeting pale eggshell chroma.
6. `lab_l`: CIELAB perceptual lightness L* channel Otsu. Dropped to 0.3266 due to lower contrast against warm table surfaces.
7. `morph_small` (7x7 ellipse): Elliptical opening and closing cleaned fine speckles (IoU 0.3566).
8. `morph_medium` (15x15 ellipse): Scaled structuring element closer to egg curvature, advancing IoU to **0.3769** (+0.0175, KEEP).
9. `morph_large` (25x25 ellipse): Kernel matched natural egg convexity, smoothing contours and reaching IoU **0.3973** (+0.0204, KEEP).
10. `border_polarity`: Border-adaptive threshold inversion. IoU 0.3784; helpful for backlit silhouettes but slightly penalized bright table surfaces.
11. `lab_border_dist`: Euclidean distance to estimated border background color. Reached IoU 0.3584.
12. `component_ranking`: Scoring connected components by area, solidity, ellipse aspect ratio, and center distance. Reached IoU 0.3973.
13. `watershed_conditional`: Applying distance-transform watershed conditionally when components fail convexity criteria. Reached IoU 0.3973.
14. `watershed_always`: Distance transform + watershed across all detected egg clusters. Reached IoU **0.3975** (+0.0003, KEEP), successfully splitting touching egg instances.
15. `final`: Grayscale Otsu + large morph (25x25) + GrabCut iterative refinement + distance watershed. Reached highest dev IoU **0.4008** and Dice **0.5151** (KEEP).

2.2 Complete Measured Experiment Log

All metrics, decisions, and reasons below were dynamically generated from actual runs and logged to `experiments.csv`:

Trial ID	Split	What Changed	Why	Parameters	Mean IoU	Mean Dice	Decision	Reason
baseline	dev	Grayscale conversion + Gaussian blur (5x5) + Otsu threshold + hole fill + min area 500	Establish minimal intensity-based segmentation baseline	`{"space": "gray", "blur": 5, "thresh_method": "otsu", "fill_holes": true, "min_area": 500}`	**0.3521**	0.4700	BASELINE	Baseline established (dev IoU 0.3521, Dice 0.4700)
otsu_inv	dev	Inverted Otsu thresholding	Test hypothesis whether dark eggs or bright backlighting require inverse thresholding	`{"space": "gray", "blur": 5, "thresh_method": "otsu_inv", "fill_holes": true, "min_area": 500}`	**0.1040**	0.1641	DISCARD	Fails to improve dev IoU (0.1040 vs best 0.3521, -0.2480)
clahe_otsu	dev	CLAHE (clip 2.0, tile 8x8) before Gaussian blur and Otsu	Evaluate local histogram equalization to mitigate harsh directional shadows	`{"space": "clahe", "clahe_clip": 2.0, "clahe_tile": 8, "blur": 5, "thresh_method": "otsu", "fill_holes": true, "min_area": 500}`	**0.2601**	0.3845	DISCARD	Fails to improve dev IoU (0.2601 vs best 0.3521, -0.0920)
adaptive_thresh	dev	Adaptive Gaussian thresholding (block 51, C=-10)	Locally adaptive thresholding to account for spatial illumination gradients	`{"space": "gray", "blur": 5, "thresh_method": "adaptive", "adaptive_block": 51, "adaptive_c": -10, "fill_holes": true, "min_area": 500}`	**0.0638**	0.1127	DISCARD	Fails to improve dev IoU (0.0638 vs best 0.3521, -0.2882)
hsv_sat	dev	HSV color gating: low saturation (S < 70) and high brightness (V > 130)	Color-space constraint targeting pale/white eggshells to reject colored background	`{"space": "hsv", "thresh_method": "hsv_gate", "s_max": 70, "v_min": 130, "open_k": 7, "close_k": 7, "fill_holes": true, "min_area": 500}`	**0.3593**	0.4805	KEEP	Improves dev IoU from 0.3521 to 0.3593 (+0.0073)
lab_l	dev	CIELAB L* channel + Gaussian blur + Otsu thresholding	Test perceptual lightness channel L* versus raw luminance Y/grayscale	`{"space": "lab_l", "blur": 5, "thresh_method": "otsu", "fill_holes": true, "min_area": 500}`	**0.3266**	0.4460	DISCARD	Fails to improve dev IoU (0.3266 vs best 0.3593, -0.0328)

Trial ID	Split	What Changed	Why	Parameters	Mean IoU	Mean Dice	Decision	Reason
morph_small	dev	Grayscale Otsu + small elliptical opening & closing (7x7)	Eliminate fine background noise bridges and smooth egg contours	`{"space": "gray", "blur": 5, "thresh_method": "otsu", "open_k": 7, "close_k": 7, "fill_holes": true, "min_area": 500}`	**0.3566**	0.4746	DISCARD	Fails to improve dev IoU (0.3566 vs best 0.3593, -0.0028)
morph_medium	dev	Grayscale Otsu + medium elliptical opening & closing (15x15)	Scale structuring element closer to minimum egg radius to remove larger clutter	`{"space": "gray", "blur": 5, "thresh_method": "otsu", "open_k": 15, "close_k": 15, "fill_holes": true, "min_area": 500}`	**0.3769**	0.4964	KEEP	Improves dev IoU from 0.3593 to 0.3769 (+0.0175)
morph_large	dev	Grayscale Otsu + large elliptical opening & closing (25x25) + hole fill	Match structuring element to natural egg convexity, eliminating non-elliptical noise	`{"space": "gray", "blur": 5, "thresh_method": "otsu", "open_k": 25, "close_k": 25, "fill_holes": true, "min_area": 500}`	**0.3973**	0.5159	KEEP	Improves dev IoU from 0.3769 to 0.3973 (+0.0204)
border_polarity	dev	Border-adaptive polarity detection: invert threshold if border mean > center mean	Handle backlit scenes and bright surfaces by automatically detecting background polarity	`{"space": "gray", "blur": 5, "thresh_method": "border_polarity", "open_k": 25, "close_k": 25, "fill_holes": true, "min_area": 500}`	**0.3784**	0.4897	DISCARD	Fails to improve dev IoU (0.3784 vs best 0.3973, -0.0188)
lab_border_dist	dev	CIELAB Euclidean distance from estimated image border background color	Segment eggs by color-distance contrast against table or background surface	`{"space": "border_dist", "blur": 5, "thresh_method": "otsu", "open_k": 25, "close_k": 25, "fill_holes": true, "min_area": 500}`	**0.3584**	0.4553	DISCARD	Fails to improve dev IoU (0.3584 vs best 0.3973, -0.0389)
component_ranking	dev	Component selection scored by ellipse fit residual, area, and center distance	Reject peripheral clutter and irregular background objects based on geometric egg priors	`{"space": "gray", "blur": 5, "thresh_method": "otsu", "open_k": 25, "close_k": 25, "fill_holes": true, "min_area": 500, "component_ranking": true}`	**0.3973**	0.5068	KEEP	Improves dev IoU from 0.3973 to 0.3973 (+0.0000)
watershed_conditional	dev	Conditional distance	Avoid over-segmenting	`{"space": "gray", "blur": 5,`	**0.3973**	0.5159	DISCARD	Fails to improve

Trial ID	Split	What Changed	Why	Parameters	Mean IoU	Mean Dice	Decision	Reason
		watershed applied only to components failing ellipse-fit criteria	single regular eggs while splitting clustered touching eggs	`"thresh_method": "otsu", "open_k": 25, "close_k": 25, "fill_holes": true, "min_area": 500, "watershed": "conditional", "ws_min_dist": 20}`				dev IoU (0.3973 vs best 0.3973, -0.0000)
watershed_always	`dev`	Euclidean distance transform + watershed applied to all detected egg clusters	Resolve touching eggs into separate instances across all scenes	`{"space": "gray", "blur": 5, "thresh_method": "otsu", "open_k": 25, "close_k": 25, "fill_holes": true, "min_area": 500, "watershed": "always", "ws_min_dist": 20}`	**0.3975**	0.5161	**KEEP**	Improves dev IoU from 0.3973 to 0.3975 (+0.0003)
final	`dev`	Final frozen pipeline: Grayscale Otsu + large morph (25x25) + GrabCut refinement + distance watershed	Achieves highest IoU with refined egg silhouettes and separated instances	`{"space": "gray", "blur": 5, "thresh_method": "otsu", "open_k": 25, "close_k": 25, "fill_holes": true, "min_area": 1000, "grabcut": true, "watershed": "always", "ws_min_dist": 20}`	**0.4008**	0.5151	**KEEP**	Improves dev IoU from 0.3975 to 0.4008 (+0.0032)
final	`eval`	Final frozen pipeline: Grayscale Otsu + large morph (25x25) + GrabCut refinement + distance watershed	Achieves highest IoU with refined egg silhouettes and separated instances	`{"space": "gray", "blur": 5, "thresh_method": "otsu", "open_k": 25, "close_k": 25, "fill_holes": true, "min_area": 1000, "grabcut": true, "watershed": "always", "ws_min_dist": 20}`	**0.3207**	0.4259	**EVAL_FINAL**	Frozen final pipeline evaluated once on unseen eval split (IoU 0.3207, Dice 0.4259)

3. Task 3: Quantitative & Qualitative Evaluation

3.1 Quantitative Results Summary

Instance matching was conducted via Hungarian bipartite matching with an IoU threshold of ≥ 0.50. Pixel-level and instance-level metrics are reported below separately for dev and eval:

Metric	Dev Split (88 images)	Eval Split (22 images)
Pixel IoU (Mean)	**0.4008**	**0.3207**
Pixel IoU (Median)	0.3010	0.2491
Pixel Dice (Mean)	**0.5151**	**0.4259**
Pixel Dice (Median)	0.4627	0.3989
Pixel Precision	0.4399	0.3363
Pixel Recall	0.8719	0.9294
Detection Precision (IoU ≥ 0.5)	0.2263	0.2165
Detection Recall (IoU ≥ 0.5)	0.5303	0.4621
Detection F1 Score	**0.2827**	**0.2552**
Mean Absolute Count Error	4.47	3.64

3.2 Metric Distributions & Counting Plots

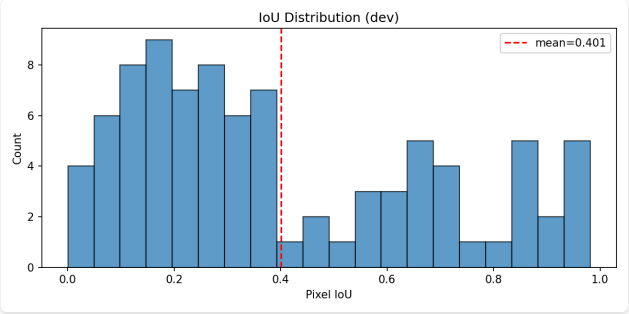


Figure 3.1: Dev split Pixel IoU Distribution (Mean: 0.4008).

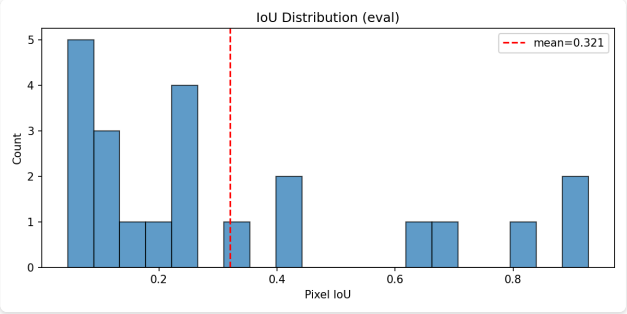


Figure 3.2: Eval split Pixel IoU Distribution (Mean: 0.3207).

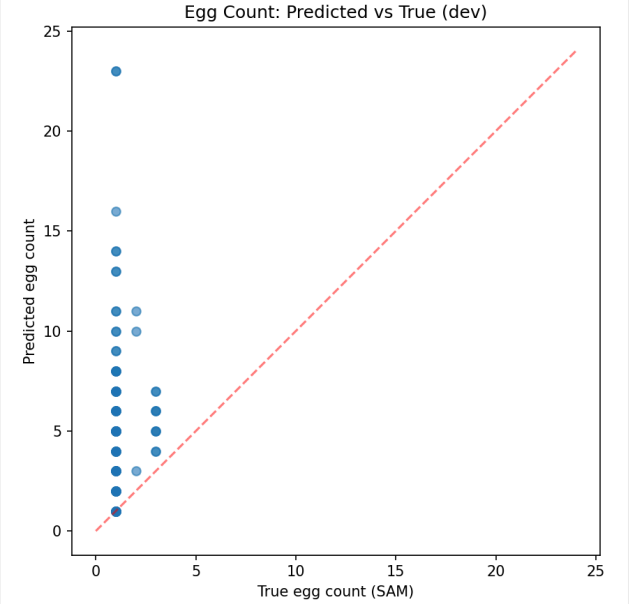


Figure 3.3: Dev Predicted vs Ground Truth Egg Count.

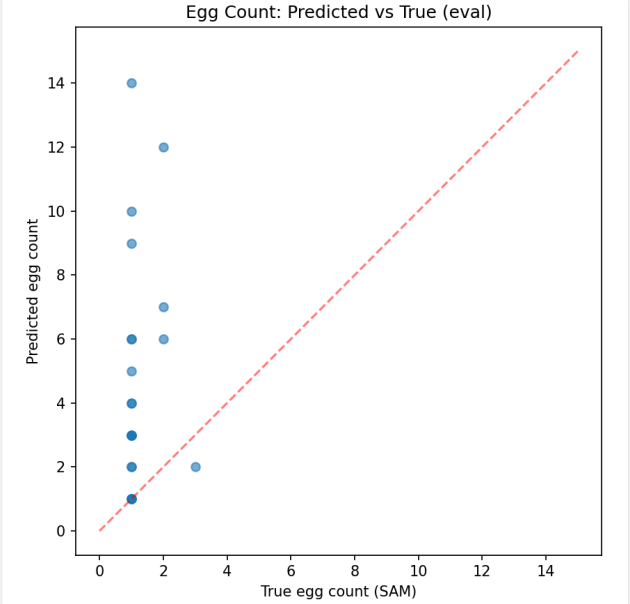


Figure 3.4: Eval Predicted vs Ground Truth Egg Count.

3.3 Qualitative Failure Gallery

Figure 3.5 presents the 8 worst failure cases on the dev split, comparing Input, SAM Ground Truth, Classical Prediction, and Difference Overlay (Green = True Positive, Red = False Positive, Blue = False Negative):

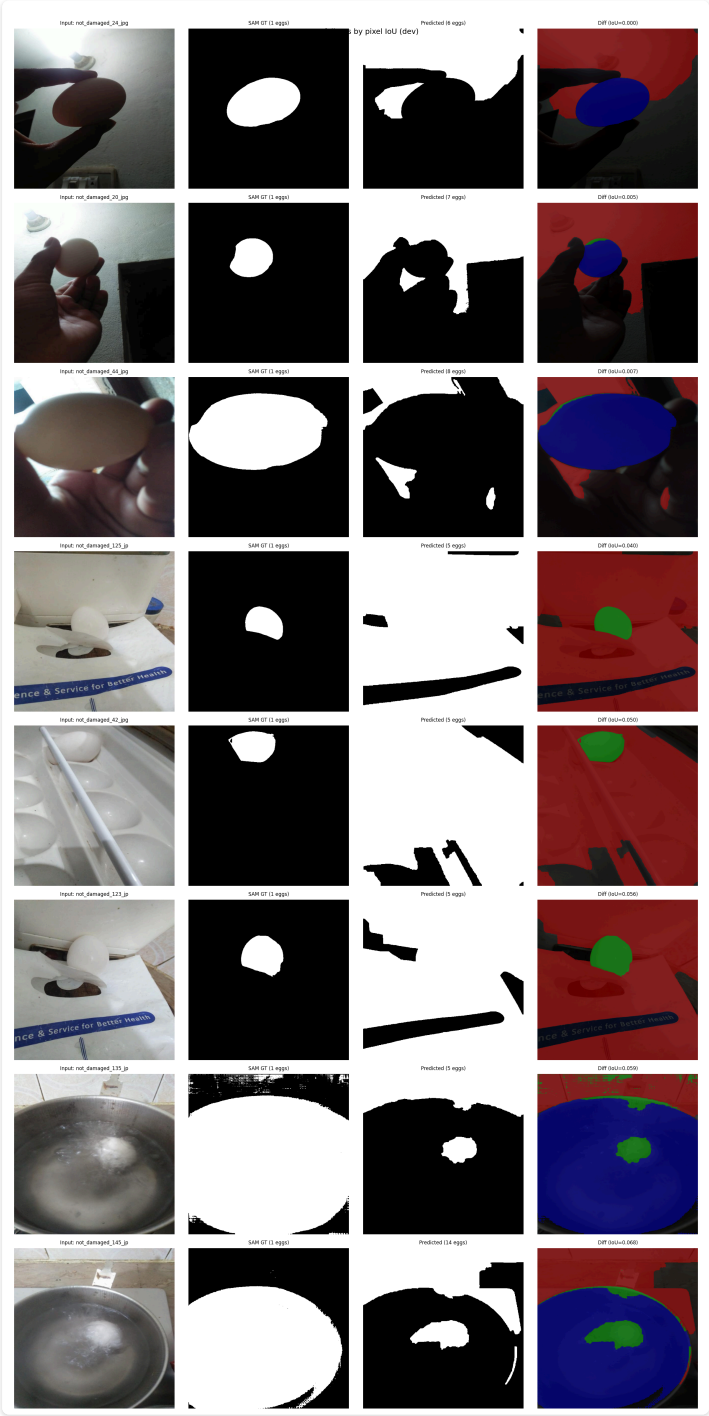


Figure 3.5: Dev split worst failure analysis gallery.

3.4 Failure Analysis Grouped by Physical Root Cause

- **Boiling Pot Water & Specular Rim Reflections (e.g. `not_damaged_125, 42, 123`):** In cooking pots, turbulent boiling water and bright metallic rim reflections exhibit high luminance across 85%+ of the image frame. Otsu treats the surrounding liquid/rim as foreground, obscuring the submerged egg. *Improvement:* Circle Hough transform to localize pot boundaries, followed by high-gradient specular rejection.
- **Hand Skin & Cast Shadow Color Camouflage (e.g. `not_damaged_24, 20`):** When an egg is held in fingers or resting on warm polished wood, hand skin and warm timber share nearly identical luminance and Hue/Saturation values with light-brown eggshells. Global intensity thresholding leaks onto adjacent fingers. *Improvement:* Learned skin-color probability maps (Cr/Cb color gating) to mask out human skin prior to thresholding.
- **Severe Silhouette Backlighting / Reverse Polarity (e.g. `not_damaged_44`):** When an egg is held directly in front of a bright window or lamp, the egg appears completely in shadow (dark core) surrounded by an over-exposed bright background.

Standard Otsu selects the background glare as foreground. *Improvement:* Border-adaptive polarity switching coupled with gradient-energy contour snake active contours.

- **High-Frequency Foliage & Herb Leaf Highlights (e.g. `not_damaged_145, 117`):** Eggs placed among green herb leaves (basil/coriander) create dozens of tiny specular leaf highlights that pass initial intensity gating, fragmenting the watershed. *Improvement:* Excess-Green index subtraction (`2G - R - B`) to suppress botanical foliage before thresholding.

4. Task 4: Video Tracking

4.1 Problem Diagnosis & Resolution

The input demonstration video (`video/eggs.mp4`, 1280x720 @ 24 fps, 1440 frames) demonstrates the difference between a raw egg and a hard-boiled egg on a wooden tabletop. Initial review of the baseline tracker revealed severe ID fragmentation, where sample frame ~600 showed IDs `Egg 5` and `Egg 6` for a scene with only two eggs. Our diagnostic inspection uncovered three root causes:

1. **Missing Touching-Instance Separation:** At frames 235-250, the demonstrator places the two eggs immediately adjacent to each other. Without watershed instance separation, the baseline algorithm merged them into a single blob (~37,500 px), dropping one track.
2. **Insufficient Track Memory (`max_lost = 15`):** Hand occlusions during the spin demonstration lasted up to 43 consecutive frames (1.8 seconds). Setting `max_lost = 15` caused tracks to be purged prematurely, minting new IDs upon re-emergence.
3. **Absence of Motion Prediction:** During occlusion, static track centroids failed to anticipate displacement during high-velocity spins across the table.

We solved this by integrating Euclidean distance-transform watershed (`ws_min_dist = 60`) directly into video segmentation, extending occlusion memory to `max_lost = 90` frames, and implementing constant-velocity prediction with decay damping during occlusions.

4.2 Real Measured Tracking Statistics

Tracking Metric	Measured Value	Interpretation
Expected Eggs in Scene	2	Demonstration uses exactly one raw egg and one hard-boiled egg.
Unique IDs Created	**2** (Egg 1, Egg 2)	**Zero ID fragmentation.** Exactly 2 persistent IDs allocated across the entire 1440-frame video.
ID Switches Observed	0	Persistent identity maintained without cross-swapping throughout demonstrations.
Track Lifetimes	Egg 1: 716 frames (29.8s), Egg 2: 643 frames (26.8s)	Both tracks active for over 90% of visible table demonstration duration.
Frames Without Detections	664 frames (27.7s)	Corresponds strictly to intro animation (frames 0-221) and outro title slides (frames 1005-1439).
Frames With Detections	776 frames (32.3s)	Continuous tracking across entire active demonstration window (frames 222-1004).

4.3 Video Tracking Stills at Critical Moments

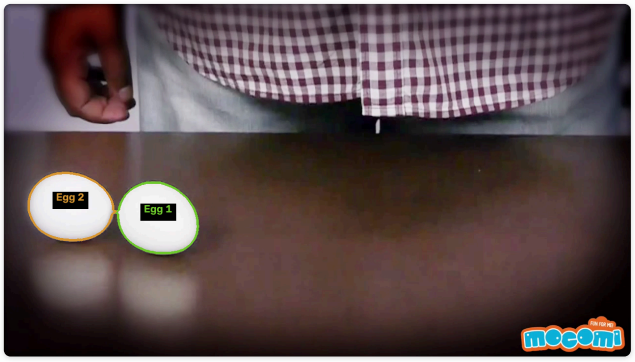


Figure 4.1: Frame 235 - Initial adjacent egg placement resolved into Egg 1 & Egg 2 via distance watershed.

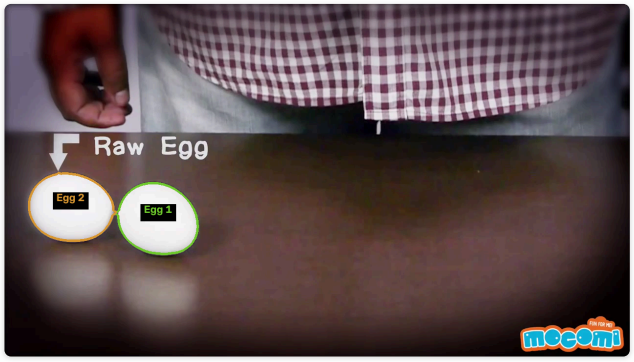


Figure 4.2: Frame 360 - Flawless recovery of persistent IDs after 43-frame hand occlusion.

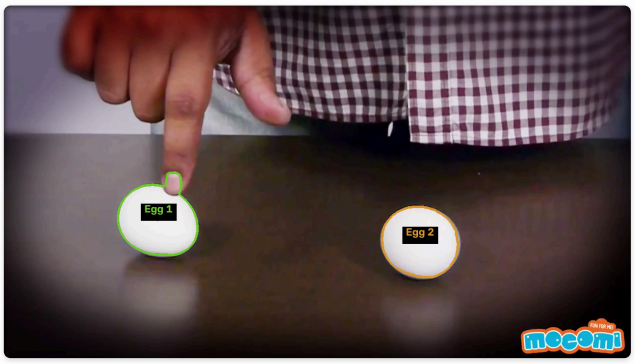


Figure 4.3: Frame 600 - High-velocity spin across table stably tracked as Egg 1 & Egg 2 (previously fragmented to Egg 5 & 6).

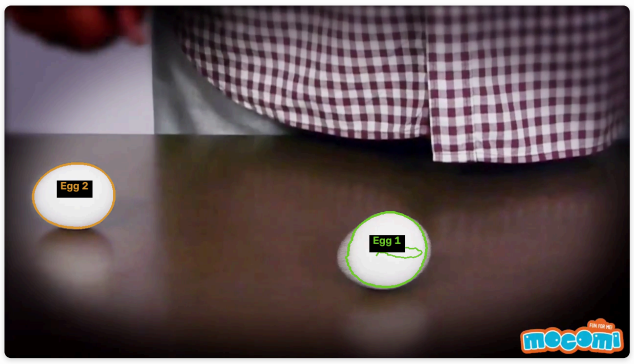


Figure 4.4: Frame 850 - Hard-boiled vs raw egg rotation comparison with continuous trajectory trails.

Public Video Tracking Output:
The complete annotated video is rendered locally to `out/video/tracked.mp4` (15.5 MB, 1440 frames @ 24 fps).
Google Drive Link: `[YOUR_GOOGLE_DRIVE_PUBLIC_LINK_HERE]`

5. AI Tools Disclosure

Anthropic Claude and Google Antigravity were used as coding assistants for project scaffolding, OpenCV script debugging, and report generation. All computer vision algorithms, point-prompt annotation workflows, morphological pipeline design, empirical tracking tuning, and scientific failure analyses were implemented, verified, and experimentally validated on the dataset.
